

Harish Inavolu

Data Analyst

+1 (940) 208-6231 | harish.inavolu31@gmail.com | linkedin.com/in/harishinavolu

Professional Summary:

Insight-driven Data Analyst with **4+ years** of experience delivering AI-enhanced analytics and data-driven solutions across healthcare domains in India and the USA. Expert in predictive modeling, machine learning, and big data analytics using Python, SQL, and cloud-native platforms. Proven ability to translate complex healthcare data into actionable insights driving operational efficiency, patient outcomes, and financial performance. Adept at building scalable ETL pipelines, automating BI dashboards, and leading cross-functional teams under Agile methodologies.

Technical Skills:

- **Programming & Databases:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Keras), SQL, R, MySQL, PostgreSQL, Oracle, SQL Server
- **AI & ML Tools:** AWS SageMaker, Azure ML Studio, Google Vertex AI, AutoML, NLP (spaCy, NLTK), Generative AI APIs (OpenAI GPT), Time Series Forecasting
- **Big Data & Cloud:** AWS (EC2, S3, Redshift, Glue, Lambda), GCP BigQuery, Azure Synapse Analytics, Hadoop, Spark, PySpark, Databricks
- **Visualization & BI:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, Streamlit
- **Data Engineering & DevOps:** Apache Airflow, Kafka, ETL Pipelines, Azure Data Factory, SSIS, Docker, Jenkins, Linux, Git, Jupyter Notebooks
- **Analytics Methods:** Predictive Modeling, A/B Testing, Statistical Inference, Data Mining, CRISP-DM, Agile/Scrum, KPI Dashboards, Data Governance & Compliance (HIPAA)

Professional Experience:

UnitedHealth Group | Dallas, TX

Jul 2023 – Present

Data Analyst

- Automated claims and reimbursement dashboards using Tableau and Power BI, reducing weekly manual reporting effort by 30%, enabling faster executive decision-making.
- Developed Python-based machine learning models predicting claim denials with 92% precision, improving financial forecasting accuracy by 25%.
- Conducted AI-powered sentiment analysis on patient feedback, identifying 3 key improvement areas that enhanced satisfaction scores by 18%.
- Engineered scalable, HIPAA-compliant ETL pipelines leveraging AWS Glue and Redshift, boosting healthcare data ingestion efficiency by 40%.
- Collaborated with clinical teams to design predictive models for chronic disease trends, supporting proactive outreach and decreasing readmissions by 12%.
- Implemented anomaly detection techniques on provider billing data, enabling early fraud detection and reducing suspected fraudulent claims by 20%.
- Developed data governance dashboards integrating multiple EMR systems, improving data reliability across teams by 25%.
- Mentored a team of 3 analysts on predictive modeling and BI best practices, increasing project delivery speed by 15%.
- Partnered with AI research to pilot a generative AI solution automating patient report summaries, saving clinicians 5+ hours weekly.

Environment: Python, SQL, AWS (Glue, Redshift, S3, Lambda), Tableau, Power BI, TensorFlow, NLP libraries, PySpark, Docker, Jira, Agile.

Apollo Hospitals | Hyderabad, India

Jan 2021 – Dec 2022

Data Analyst

- Automated clinical KPI dashboards using Power BI and SQL, cutting manual tracking time by 40% and enhancing executive reporting accuracy.
- Built predictive Python models forecasting hospital admissions, increasing resource allocation efficiency by 20%.
- Consolidated patient data across multiple branches via Airflow pipelines and MySQL, achieving 99% data accuracy and availability.
- Conducted statistical analysis on EHR data to identify treatment success factors, improving patient outcomes by 10%.
- Collaborated with physicians to develop ML risk stratification models, enabling early intervention for chronic disease management.
- Optimized ETL workflows on Azure Data Factory, reducing enterprise dashboard data refresh times from 6 hours to under 1 hour.
- Created NLP scripts for extracting structured clinical terms from unstructured notes, enhancing analytics readiness by 25%.
- Employed AI-driven anomaly detection to analyze medication error root causes, reducing error incidents by 18%.
- Led migration of legacy Excel reports to standardized Power BI dashboards used across 12 departments.

Environment: Python, SQL, Power BI, Azure Data Factory, MySQL, Apache Airflow, NLP (spaCy, NLTK), Pandas, Matplotlib, Docker, Linux.

Educational Qualification:

- M.S. in Data Science – University of North Texas, Denton, TX
- B.Tech in Electronics & Communication Engineering – Narasaraopeta Engineering College, India